

MAROUANE BEN HADDOU

DATA ENGINEERING STUDENT

Address: Centre des Travaux Agricoles 3003 Targuist-Al Hoceima, MOROCCO
Phone: +212 6 55 18 44 02
Email: marouanebenhaddou9@gmail.com
LinkedIn: linkedin.com/in/marouane-ben-haddou
Portfolio: marouane-benhaddou.vercel.app

SUMMARY

Final-year Data Engineering student at ENSAH, seeking **job opportunities or a final-year internship (PFE)** in Data Engineering, Big Data or Artificial Intelligence. Hands-on experience building data pipelines, AI agents and software platforms within engineering teams.

PROFESSIONAL EXPERIENCE

AI SOFTWARE ENGINEER — GOJI — Sep. 2026 – Present

Paris, France (Remote)

- Continued working with Goji after the internship, developing the Hospitality agent to handle guest requests and support hotel workflows.
- Tested and evaluated the Goji Sales agent on customer use cases to assess its responses and behavior.

SOFTWARE ENGINEERING INTERN — GOJI — Jul. 2026 – Sep. 2026

Paris, France (Remote)

- Helped design and develop AI agents for personalized product exploration and hotel concierge services.
- Built the Hospitality inbox service for guest and hotel staff communication.
- Developed concierge features for guest requests, hotel and reservation context retrieval, and connected workflows.
- Built a CI/CD pipeline and implemented evaluations for Goji's AI sales agent.
- Used Python, FastAPI, PostgreSQL, pgvector, ChromaDB, the Anthropic SDK and the WhatsApp API.

AI ENGINEER INTERN — SMART AUTOMATION TECHNOLOGIES — Jul. 2026 – Sep. 2026

Tangier, Morocco (Remote)

- Built a multi-agent AI platform for warehouse operations and forecasting, with parallel execution, conflict detection and constraint validation.
- Combined SQL, vector search and BM25 to ground recommendations in warehouse data and procedures.
- Developed a React/TypeScript UI and JWT-secured FastAPI backend; designed warehouse adapters and local LLM inference.
- Built the system using Python, LangGraph, Ollama, FastAPI, React, TypeScript, PostgreSQL, Qdrant, Redis and Docker.

DATA ANALYST INTERN — DPA (PROVINCIAL DEPARTMENT OF AGRICULTURE) — Jul. 2025 – Aug. 2025

Al Hoceima, Morocco (On-Site)

- Collection and processing of climate and agricultural data, exploratory data analysis (EDA), integration into Snowflake and Power BI dashboard for business stakeholders.
- Used Python and SQL for data processing, Snowflake for storage and Power BI for reporting.

PROJECTS

TAASIM — REAL-TIME URBAN MOBILITY PLATFORM — STREAMING / KAPPA

- Kappa architecture: 1.7M Porto trips remapped via OSRM onto Casablanca and replayed as a real-time Kafka stream.
- Built three event-time Flink jobs for GPS normalization, supply/demand aggregation and trip matching, with MinIO checkpoints and Cassandra state.
- Forecast demand by zone in 30-minute windows using Spark MLlib, reducing RMSE by 22% versus baseline; served results through FastAPI and Grafana.
- **Tools:** Kafka, Flink, Spark, Cassandra, MinIO, OSRM, Grafana, FastAPI, Docker

PROCUREMENT SYSTEM BIGDATA PIPELINE — BIG DATA

- ETL pipeline orchestrated by Apache Airflow (8-task daily DAG) processing orders, inventory and master data.
- Distributed HDFS storage, Hive warehouse and Cassandra snapshots, queried through multi-source Trino federated queries.
- Automatic net demand calculation (safety stock) and automated generation of optimal supplier orders.
- **Tools:** Apache Airflow, Hadoop HDFS, Apache Hive, Apache Cassandra, Trino, PostgreSQL, Docker

FINDJOB — JOB RECOMMENDATION PLATFORM — DATA ENGINEERING / AI

- Multi-source aggregation (Adzuna, ReKrute, Emploi-Public via Selenium + OCR) in a medallion architecture MongoDB and PostgreSQL (star schema).
- Built multilingual semantic search with pgvector, ranked by seven signals, plus NLP-based CV parsing and ML salary estimation.
- Asynchronous Kafka processing (CV parsing) + Redis cache (sub-second API), React front-end and a Power BI job-market dashboard.
- **Tools:** FastAPI, React, PostgreSQL/pgvector, MongoDB, Kafka, Redis, Airflow, Power BI, Docker

EDUCATION

- **National School of Applied Sciences (ENSAH) Al Hoceima** — **2024 – Present**
  Engineering Degree in Data Engineering
- **National School of Applied Sciences (ENSAH) Al Hoceima** — **2022 – 2024**
  Integrated Preparatory Cycle

ADDITIONAL INFORMATION

- **Languages & Frameworks:** C, Python, Java, PL/SQL, FastAPI, React, Spring Boot
- **Databases:** Oracle, MySQL, PostgreSQL, MongoDB, Snowflake, Cassandra, Redis
- **Big Data & ETL:** Hadoop, Airflow, Hive, Trino, Flink, Kafka, MinIO
- **Machine Learning:** Scikit-learn, Pandas, PyTorch, TensorFlow, MLflow
- **Generative AI:** LangChain, LangGraph
- **Scraping & Viz:** BeautifulSoup, Playwright, Power BI
- **DevOps:** Git, GitHub, Docker, Grafana, Prometheus
- **Methodologies & Design:** Agile/Scrum, UML, Merise
- **Languages:** Arabic (Native), French (C1-TCF), English (Advanced), Spanish (Basic)